Vision and Scope Document

for

Plant

Version 1.0 approved

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8/27/26

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Revision History

Name	Date	Reason For Changes	Version
Plant	8/27/26	*Created document*	1

1 Business Requirements

1.1 Background

Existing social media platforms tend to specialize narrowly: Reddit organizes discussion around static, pre-existing communities; Discord enables real-time community spaces but requires users to already know where to look; Tumblr emphasizes personal expression but lacks structured discussion. Users interested in a niche topic often can't find (or build) a dedicated community until that community already has enough critical mass to justify a dedicated server or subreddit. No platform lets a single post organically grow into its own community space based on the interest it generates.

1.2 Business Opportunity

Plant addresses this gap by allowing any post to evolve into its own forum once it reaches a popularity threshold – removing the barrier between posting content and building a community around that content. This lowers the cost of community formation and lets user interest, rather than upfront organizational effort, determine which topic gets dedicated spaces.

1.3 Business Objectives and Success Criteria

The business objective of Plant is to create a social media platform that has the potential to attract and retain users. This is especially critical during the initial release of the software to start producing a measurable return on investment. Due to this, the most important success metrics for Plant to track is users. Other important success metrics worthy of being considered include the number of posts that “graduate” into forums per week, the average lifespan/retention after a forum “spawns”, user engagement rates (posts, comments, votes) within spawned forums vs. original posts, and growth rate of active daily/monthly users. Additionally, user satisfaction has a direct correlation to user retention. Therefore,

while not as important, quality survey results of users in 3-month intervals after release would be beneficial to track.

When it comes to the financial metrics of Plant, there are several facets to explore; one of these facets is running ads. There would be two tiers to a user/entity running an ad on our platform, which include:

- **Personal Ads** – a single user can pay a price to have their post/forum advertised on the explore page, with the range of advertisement reach being variable based on the ad running price they choose to pay.

- **Business Ads** – a company or entity can go through the same process as a single user to run advertisements on our platform, with additional steps, including their inclusion of the type of business/good/service they are advertising, their target audience, and desired customer demographics, which leads into a second financial metric.

Our second financial metric involves selling user data that only aligns with business demographics that exclude personal information and data. This includes:

- Telephone numbers
- Addresses
- Emails
- Answers to security questions
- Passwords used for every individual account

Ultimately, we offer data that ensures the safety of our users while also appealing to businesses who are interested in garnering customers through our platform. This data entails the following:

- The communities/forums a user visits.
- Keystrokes recorded from the community/forum query bar.

- Rough geolocational data that is updated every 36 hours from a user opening the app starting at 12:00 am on any given day.
- Posts that users interact with (liking, commenting, saving).

We encourage customization of an individual's profile. By offering our users a premium service such as Discord's "Nitro" subscription. Our users can gain access to premium customization options such as user flairs, animated profile backgrounds, and unique user tags. This service would also limit advertisements to appear only on the explore feed, disabling them from being seen inside forums and communities.

1.4 Customer or Market Needs

When it comes to social media applications, portability, or the ability to run the software almost anywhere on anything, is one of the top priorities for users. Additionally, users expect a seamless transition between features and dashboards as well as an interactive, user-friendly interface. As for the content presented to the user, posts and forums need to be able to grow naturally without the influence of paid-for features. Content that is presented should fit the user's preferences while still providing an equal opportunity for other posts to grow. Because of these needs and expectations Plant has the following interface and performance requirements:

1. The application must be able to work as a web app and run with minimal hardware requirements.
2. Pages need to respond quickly and update with life, accurate information.
3. The interface must be intuitive enough for new users to feel comfortable while also promoting personalization of accounts and look of feed.
4. The interface must include a customizable, interactive dashboard that allows users to pin posts/forums and adjust the layout to show the content they want.

1.5 Business Risks

Developing a social media application entails some associated business risks, most notably competition from marketplaces like Reddit and Stack Overflow. Additionally, gaining user attention and retention would be difficult given the level of competition. To mitigate some of this risk, it would be best to release the app with the core major features and implement new features incrementally. Following this plan, surveys can be presented to users requesting suggestions for desired features or improvements. Other acknowledged business risks are outlined below:

- Moderation risk – auto-spawned forums could proliferate faster than moderation capacity.
- Fragmentation risk – content could scatter into too many small forums, reducing engagement in any specific one.
- Technical risk – determining a fair, gaming-resistant “popularity threshold” algorithm.
- Adoption risk – users accustomed to existing platforms may not understand the post-to-forum mechanic without onboarding.

2 Vision of the Solution

2.1 Vision Statement

For social media users who want community spaces to form naturally around shared interests, Plant is a social forum platform that lets popular posts grow into their own dedicated forums. Unlike Reddit, Discord, or Tumblr, which require communities to be

manually created before content can gather around them, Plant lets community formation emerge organically from what users are already engaging with.

2.2 Major Features

1. Post progression to forum through an interactivity score.
2. Personal account creation and customization.
3. Post/forum query bar.
4. Explore page.
5. Advertising/business account creation and customization
6. Favorites/Recent pages.
7. Individual profile pages.
8. Direct messaging between users
9. Interactive, customizable dashboard for user posts and activity.

2.3 Assumptions and Dependencies

Throughout the planning process for Plant, some assumptions and dependencies became abundantly clear. Some of these assumptions made by us include: a good budget for advertising and drawing attention to the platform, access to a smaller initial database to store advertising data collected from users and posts, ability to partner and collaborate with advertising corporations, and a continued budget extension following the release of the application to allow for infrastructure and development expansion. Along with these assumptions, Plant is dependent on: creating a scoring system for promoting posts that is effective and fair, gaining and retaining users, maintaining a favorable relationship with third-party data brokers, creating a secure robust system that is resistant to attacks, ensuring data is handled properly and sensitive data is stored for the least amount of time necessary, and ensuring that previously established user data collection policy is followed and does not violate users privacy.

3 Scope and Limitations

The purpose of Plant is to create a social media platform that users enjoy interacting with and provides post equal opportunities to grow and develop communities around them without the influence of organizational funding or backing. To achieve this, features that promote artificial growth or interactivity with post fall outside of the scope of this project and will be rejected. Additionally, preventive measures to prevent artificial growth will be limited to detection of bot accounts to boost interactivity. The features delivered in the first three planned batches of releases are limited to the previously defined features; any addition of new features in those batches will fall out of scope due to budgeting and time restraints. The scheduling of planned release times cannot exceed a delay of up to a month before falling out of project scope. Additionally, development and infrastructure that will cause costs to exceed 15% over the estimated budget will consequently fall out of scope.

3.1 Scope of Initial Release

In the initial release of Plant, Features will be limited to the core functionality features that users expect from a social media app and the post-to-forum progression. During this release period Plant's focus will be on gaining users and generating income through advertising data collection and distribution. By limiting the initial release, development cost will be lowered by allowing for a smaller team of developers while also providing time to polish the user interface and core features that will attract new users. For the initial release period, changes are possible and likely to happen, therefore, development will allow for changes to be made to the defined features. However, the core features included in the initial release should not be changed unless deemed necessary. The core features that Plant will initially be released with include:

1. Post-to-forum progression.
2. Explore page.
3. Personal account management.
4. Direct messaging.

5. Some profile customization features (profile pictures, bios, recent activity).

3.2 Scope of Subsequent Releases

After the initial release, development focus will shift to the second batch of features as well as maintaining the core features. At this point, the user base will be better established, allowing advertising to become feasible. This will allow for a secondary route of income and provide for more costs to be directed to development and infrastructure. Similarly to the conditions set for the first release, changes to the way features function are expected, however, changes to the planned release features should not be made. The features planned for the first batch include:

1. Post/forum query bar.
2. Individual profile showcase pages (full range of customization)
3. Monetization of profile cosmetics through premium Plant's subscription service.
4. Advertising Account management with tiers
5. Maintaining initial release through patches, bug reporting, and database management.

Following the secondary batch of features, the results from the first user satisfaction survey will be accounted for, and release of the planned final batch will begin before the main shift goes to maintaining and user recommended features. This batch was purposely planned with only a few features to allow for changes to preexisting features or the user-interface. This final release batch will include:

1. Favorites/recent pages.
2. Interactive user dashboard.
3. User recommended improvements/features.
4. Patching and maintenance of already released features.

3.3 Limitations and Exclusions

Stakeholders are primarily worried about revenue and gaining a high return on investment as quickly as possible. Because of this, they may anticipate features such as paid post promotion or paying to promote certain posts to grow into forums. This goes conversely against our vision statement of creating a platform that allows post naturally propagate and grow or die. Additionally, stakeholders might anticipate paid users to still receive targeted ads. However, while being an incentive to upgrade, this also provides a higher quality of life for premium users, providing more reason for their continued support.

4 Business Context

Plant's business model consists of data distribution, a subscription-based service, and advertisement hosting through business accounts. Given that Plant is a social media app a few challenges expected to appear are standing out from rival competition, convicting users to switch from more familiar platforms, and maintaining user-centered data collection policies that the business will be able to use for advertising data. The expected customer categories our business will service consists of:

1. Free users – These users will not be paying for the subscription service but are still major contributors to user satisfaction, data collection, and providing a place to host ads. Because of this, user's experience still needs to be enjoyable while limiting account personalization.
2. Premium users – These users will be paying Plant a continuous subscription while contributing to data collection, community building, and establishing a loyal userbase. For these reasons, users in this category will be

shown minimal to no ads, given full access to personalization options, and more quality-of-life features (premium reactions, higher quality image attachments, profile badges)
3. Business accounts or advertisers – The primary focus of these accounts is to advertise their products to users of the platform. Therefore, the account personalization will be the same as free users; however, the user will have advertising options to submit possible ads to present in the explore page. Additionally, the account will have in-depth analytics on interactivity and demographics reached.
4. Admin or developer accounts – These users need access to experimental features, administrative features (mute, ban, remove content), as well as access to a development and production pipeline. These users will also have access to the backend of Api for development.

During the conception of Plant, a few assumptions had to be made to plan for future development and cost predications. These assumptions include: the ability to gain and retain a userbase, customer feedback will not require extensive changes to plans, and development cost will be able to remain lowered will a smaller development team. From a management standpoint, the most important priorities for this project include:

1. Ensuring release goals are met with full list of features in the given time interval.
2. Ensuring the quality of the platform and user satisfaction.
3. Ensuring that production and development cost do not exceed over 15% of the budget.

4.1 Stakeholder Profiles

The prospected stakeholders for our business have been split into three profiles: executive, editors, and legal aid. The chart provided below will present their respective values, attitudes, major interest, and constraints in the project.

Stakeholder	Major Value	Attitudes	Major Interests	Constraints

Executives	increased revenue	see product as avenue to 25% increase in market share	richer feature set than competitors; time to market	Maximum budget = $1.4M
Editors	fewer errors in work	highly receptive, but expect high us-ability	automatic error correction; ease of use; high reliability	must run on low-end work-stations
Legal aides	quick access to data	resistant unless product is keystroke compatible with current system	ability to handle much larger database than current system; easy to learn	no budget for retraining

4.2 Project Priorities

The table below covers the project's major priorities with justification of each priority and their related constraints.

Dimension	Driver (state objective)	Constraint (state limits)	Degree of Freedom (state allowable range)
Schedule	release 1.0 to be available by 10/1, release 1.1 by 12/1	Time meeting with groups and confirming changes	Everyone is allowed to always work on the project with a minimum requirement of 40 hours a week
Features	The friends tree, posts, accounts,	Learning how to implement features as we go	70-80% of high-priority features must be included in release 1.0
Quality	Make sure the website is functional and cleaned up	We must make the development on top for a 1.0 release	90-95% of user acceptance tests must pass for release 1.0, 95-98% for release 1.1

Staff	Assign staff to what areas need work	maximum team size is 6 developers + 4 testers	Each group member can decide where they work
Cost	Make the app with what tools we have	We will be unlikely to buy/afford expensive coding software	Budget overrun up to 15% acceptable without executive review

4.3 Operating Environment

Early in Plants development, our first users will likely begin logging in everywhere around the globe. Widening the reach while also keeping users in the same geographical region close together will inevitably place our users in several different time zones. As outlined above in section 1.3, our users geolocational data will be updated every 36 hours beginning at 12:00AM on any given day, this will be measured based on the last recorded log-off to the next recorded login in tandem with each of our users' phones/computers internal clock data. Given the information above, our users can access the platform whenever they want, whenever they feel it is necessary. The data of our users will be generated at Content Delivery Networks (CDNs) near regional hubs to be able to deliver media files and feeds quickly to users, and by using data partitioning we will route user profiles and posts to specific regional data clusters to reduce latency. Due to the ideal global use of Plant, we will build modular consent management tools to be able to adapt to varying data collection and tracking laws per country. With that in mind, yes data from multiple locations will be combined so that we will be offering cohesive user experience worldwide, but that experience will follow laws and regulations set in place by the governments of the regions in which our users reside and will be filtered accordingly. Data access will be made available as instantaneously as possible once it is filtered through each region's modular consent management system. Continuous access is preferred, with only one system at a time going down for maintenance. If a full site shutdown is needed, a message will be sent out 48 hours beforehand. With data/content security controls

being acknowledged above with the mention of compliance with regional moderation laws, that still leaves use with protecting user's data that will be intentionally kept private and excluded from our morals of selling user data as outlined in section 1.3. The measures we will take to protect this data will include data encryption, 2FA and MFA for user login sessions, and an alert system that will notify a user of every login attempt made. We also offer users multiple options for their personal account protection including the limitation of account visibility and what is visible on their profiles.